

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION COURSE CODE: ITA0515 Slot B

COURSE OUTCOME COVERED

CO4: Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5) Assessment Tool Weightage: 40%

QUESTIONS:5 Total Marks 50 Duration :60 Minutes Annexure A

Data Interpretation

Code of Conduct:

I V. Thanusri (192472265)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: V.Thanusri

Rubrics for Evaluation

Criteria	Weight age (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks (100)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data	
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided	
Application of Concepts	20%	Effectively applies relevant concepts/tool	Applies concepts with	Limited application;	Unable to apply concepts	

		s to interpret data accurately	minor errors	requires guidance	to data	
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions	
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand	

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Structure from Motion Consistency

Two datasets produce:

- Dataset A: accurate depth but requires high-quality images
- Dataset B: inconsistent depth but works on low-quality images

Compare the datasets and evaluate which setup is more practical for real-world deployment.

Justify your reasoning.

2. Stereo Vision vs Motion Estimation

Depth estimation results:

- Stereo Vision: accurate but requires dual cameras
- Motion Estimation: less accurate but works with single camera

Compare both approaches and evaluate which is preferable under hardware constraints. Justify your choice.

3. Model Generalization vs Overfitting

Performance:

Model Training Testing

A 98% 70%

B 90% 88%

Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

4. YOLO Performance across Scales

Detection results:

- Large objects: 95%
- Medium objects: 85%
- Small objects: 60%

Compare performance across scales and evaluate whether YOLO alone is sufficient for multi-scale detection tasks. Justify improvements.

5. Motion Model Selection for Animation

Two models:

- Model A: smooth but physically inaccurate
- Model B: realistic but computationally intensive

Compare the models and evaluate which should be used for animation systems under different constraints (real-time vs cinematic). Justify your decision.

1. Comparison and Evaluation of Computer Vision Techniques

Introduction

In computer vision and machine learning applications, selecting the best technique requires more than simply looking at accuracy. A technique that gives very high accuracy may require expensive hardware, high-quality input data, or large computational resources. Similarly, a

technique with slightly lower accuracy may be more useful if it is robust, inexpensive, and works reliably in real-world conditions.

Therefore, techniques should be compared using several factors such as accuracy, robustness, computational requirements, hardware requirements, generalization ability, reliability, and suitability for the intended application. The following sections compare different techniques and justify the most appropriate choice for each situation.

1. Structure from Motion Consistency

Structure from Motion (SfM) is a computer vision technique used to recover information about the three-dimensional structure of a scene from a sequence of two-dimensional images. It can estimate both the movement of the camera and the depth or 3D structure of objects in the scene. In the given situation, two datasets produce different results. Dataset A provides accurate depth information but requires high-quality images. Dataset B produces inconsistent depth information but is capable of working with low-quality images.

However, the major limitation of Dataset A is its dependence on high-quality images. Real-world environments are not always controlled. Images may be affected by poor lighting, shadows, motion blur, camera noise, reflections, low resolution, or changes in the environment. If the input quality decreases, the performance of Dataset A may also decrease significantly. This makes it less flexible for uncontrolled environments. Dataset B has the opposite characteristics. Its depth estimation is inconsistent, which means that it may not provide reliable depth values for every part of a scene. This is a disadvantage when precise depth information is required. However, Dataset B can work with low-quality images, giving it better robustness in difficult environments.

Final Evaluation

Dataset B is more practical for general real-world deployment because of its ability to handle low-quality images and its greater robustness. Dataset A is preferable when accurate depth estimation is critical and high-quality image acquisition can be guaranteed. Therefore, the decision should depend on the application requirements and the expected quality of the input data.

2. Stereo Vision vs Motion Estimation

Stereo Vision and Motion Estimation are two important techniques used for estimating depth from images. Stereo Vision uses two cameras positioned at different viewpoints, while Motion Estimation generally uses a single camera and obtains depth information from the camera's movement across multiple frames.

Stereo Vision has an important advantage because it can provide relatively accurate depth information. Since two cameras observe the same scene from different positions, the difference between the two views can be used to estimate the distance of objects. This makes Stereo Vision suitable for applications such as robotics, autonomous vehicles, 3D reconstruction, and industrial inspection.

However, Stereo Vision requires two cameras. The cameras must be properly positioned and calibrated to obtain accurate results. The system also needs additional hardware, which increases the overall cost and complexity. If the cameras are not properly synchronized or calibrated, the depth estimation can also become inaccurate. Motion Estimation uses a single camera and estimates depth by analyzing changes between consecutive frames as the camera moves. Its main advantage is the lower hardware requirement. Only one camera is required, making the system cheaper and easier to install.

The disadvantage is that Motion Estimation generally provides less accurate depth than Stereo Vision. It may also be affected by camera movement, scene motion, lighting changes, and insufficient visual features. Therefore, it may not be suitable for applications requiring highly precise depth measurements. When hardware resources are limited, Motion Estimation becomes a better option. For example, a small robot or mobile application may not have enough space or budget for two cameras. In such situations, using a single camera provides a practical solution. On the other hand, if the system has sufficient hardware resources and requires high-quality depth information, Stereo Vision is a better choice.

Final Evaluation

Under strict hardware constraints, Motion Estimation is preferable because it requires only one camera, reduces system cost, and simplifies deployment. However, if depth accuracy is more

important than hardware cost, Stereo Vision should be selected. Thus, the choice depends on the application's priorities. Motion Estimation provides better hardware efficiency, while Stereo Vision provides better depth accuracy.

3. Model Generalization vs Overfitting

Machine learning models must be evaluated not only on their training data but also on unseen testing data. A model that performs very well on training data but poorly on testing data may have learned the training examples too specifically. This problem is known as overfitting.

The given results are:

- Model A: Training accuracy = 98%, Testing accuracy = 70%.
- Model B: Training accuracy = 90%, Testing accuracy = 88%.

At first, Model A appears to be the better model because it has a training accuracy of 98%. However, its testing accuracy is only 70%. There is a difference of 28 percentage points between the training and testing results.

This large performance gap indicates that Model A is likely overfitting. It has learned the training data extremely well but does not perform equally well when presented with new data. In a real-world application, this can be a serious problem because the model will encounter data that it has never seen during training.

For example, if a model is trained to recognize vehicles, it may achieve very high accuracy on the training images but fail when vehicles are captured from different angles, under different lighting conditions, or using different cameras. Such a model would not be reliable for deployment.

Model B has a training accuracy of 90% and a testing accuracy of 88%. The difference is only 2 percentage points. This indicates that Model B has learned patterns that generalize well to unseen

data. Although its training accuracy is lower than Model A, its testing performance is significantly better.

Generalization is especially important in real-world applications because the model needs to handle different inputs and conditions. A model with good generalization is less likely to fail when it encounters new examples.

Model B also has a lower risk of unpredictable behavior after deployment. It provides a better balance between training performance and testing performance.

Final Evaluation

Model B is more reliable for deployment. Its 88% testing accuracy is much higher than Model A's 70%, and the small gap between its training and testing accuracy indicates good generalization. Model A should not be selected simply because it has a higher training accuracy. Its large training-testing gap indicates overfitting and creates a higher risk of poor performance on real-world data.

Therefore, for deployment, testing accuracy and generalization should be given greater importance than training accuracy alone.

4. YOLO Performance Across Scales

YOLO is a popular object detection algorithm used to identify and locate objects in images and videos. One of its important advantages is its ability to perform object detection quickly. However, detection performance can vary depending on the size of the objects in the image. The given detection results are:

- Large objects: 95% accuracy.
- Medium objects: 85% accuracy.
- Small objects: 60% accuracy.

The results clearly show that YOLO performs best when objects are large. The accuracy of 95% indicates that most large objects are detected correctly. Large objects occupy more pixels in the image, which provides the model with more visual information such as shape, edges, texture,

and appearance. For medium-sized objects, the accuracy decreases to 85%. Although this is still a good result, there is less visual information compared with large objects. The detector therefore has a greater chance of missing or incorrectly classifying some objects.

The largest problem occurs with small objects. The detection accuracy is only 60%, which means that a significant number of small objects may be missed or incorrectly detected. Small objects occupy very few pixels and may lose important features when the image is resized during processing.

This is particularly important in applications such as traffic monitoring, surveillance, drone-based detection, autonomous systems, and aerial image analysis. In these applications, small objects can be important even though they occupy only a small portion of the image.

YOLO alone may therefore not be sufficient when the application requires reliable detection across multiple object scales. However, YOLO can be improved using several approaches.

A model specifically optimized for small-object detection can further improve the results.

Final Evaluation

YOLO provides excellent performance for large objects and good performance for medium objects, but its 60% accuracy for small objects indicates a significant weakness. Therefore, YOLO alone is not sufficient for highly reliable multi-scale detection when small objects are important.

It can still be used as the main detection model, but it should be combined with higher-resolution inputs, multi-scale feature extraction, suitable training data, and small-object detection improvements.

5. Motion Model Selection for Animation

Motion models are used to generate movement in computer animation, games, simulations, virtual environments, and cinematic applications. The choice of a motion model depends on the required realism, computational resources, and whether the system needs to operate in real time. In the given case, Model A produces smooth motion but is physically inaccurate. Model B produces realistic and physically accurate motion but is computationally intensive. Model A has the advantage of being computationally efficient. It can generate smooth movement without requiring

a large amount of processing power. This is particularly important in real-time applications where the animation must respond immediately to user input.

Examples include video games, interactive simulations, virtual reality applications, augmented reality systems, and real-time character animation. In such systems, users expect smooth and responsive movement. A highly realistic model that causes delays may reduce the quality of the overall experience. The main disadvantage of Model A is its physical inaccuracy. The motion may look smooth but may not behave exactly like real-world movement. For some applications, this is acceptable because visual smoothness and responsiveness are more important than physical accuracy.

Model B provides more realistic motion because it considers physical behavior more accurately. This makes it suitable for applications where realism is the primary objective. However, the model requires more computational resources and may not be able to operate efficiently in real time on limited hardware.

For example, a movie production may spend significant processing time generating a highly realistic movement because the final visual quality is more important than real-time response.

Final Evaluation

For real-time applications, Model A is preferable because it provides smooth movement with lower computational requirements. The slight loss of physical accuracy may be acceptable when fast and responsive animation is required. For cinematic applications, Model B is preferable because it provides more realistic and physically accurate movement. The higher computational cost is acceptable because cinematic rendering usually prioritizes visual quality over real-time performance.

Therefore, neither model is universally better. The correct choice depends on the application's requirements, available hardware, processing time, and required level of realism.